

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, co
from sklearn.preprocessing import LabelEncoder
import joblib

# !pip install openpyxl
# import Libraries & Load Data
# !pip install joblib

#Define the path to the Excel file
# IMPORTANT: The current path points to a local file c
file_path = r"/content/sample_data/Prediction_data.xls

# Define the sheet name to read data from
sheet_name = 'vw_ChurnData'

# Read the data from the specified sheet into a pandas
data = pd.read_excel(file_path, sheet_name=sheet_name)

# Display the first few rows of the fetched data
print(data.head())

```

|   | Customer_ID | Gender | Age | Married | City         | Number_of_Referrals | \ |
|---|-------------|--------|-----|---------|--------------|---------------------|---|
| 0 | 0002-ORFBO  | Female | 37  | Yes     | Frazier Park | 2                   |   |
| 1 | 0003-MKNFE  | Male   | 46  | No      | Glendale     | 0                   |   |
| 2 | 0004-TLHLJ  | Male   | 50  | No      | Costa Mesa   | 0                   |   |
| 3 | 0011-IGKFF  | Male   | 78  | Yes     | Martinez     | 1                   |   |
| 4 | 0013-EXCHZ  | Female | 75  | Yes     | Camarillo    | 3                   |   |

|   | Tenure_in_Months | Phone_Service | Multiple_Lines | Internet_Service | ... | \   |
|---|------------------|---------------|----------------|------------------|-----|-----|
| 0 | 9                |               | Yes            | No               | Yes | ... |
| 1 | 9                |               | Yes            | Yes              | Yes | ... |
| 2 | 4                |               | Yes            | No               | Yes | ... |
| 3 | 13               |               | Yes            | No               | Yes | ... |
| 4 | 3                |               | Yes            | No               | Yes | ... |

|   | Payment_Method  | Monthly_Charge | Total_Charges | Total_Refunds | \ |
|---|-----------------|----------------|---------------|---------------|---|
| 0 | Credit Card     | 65.599998      | 593.299988    | 0.000000      |   |
| 1 | Credit Card     | -4.000000      | 542.400024    | 38.330002     |   |
| 2 | Bank Withdrawal | 73.900002      | 280.850006    | 0.000000      |   |
| 3 | Bank Withdrawal | 98.000000      | 1237.849976   | 0.000000      |   |
| 4 | Credit Card     | 83.900002      | 267.399994    | 0.000000      |   |

|   | Total_Extra_Data_Charges | Total_Long_Distance_Charges | Total_Revenue | \ |
|---|--------------------------|-----------------------------|---------------|---|
| 0 | 0                        | 381.510010                  | 974.809998    |   |

|   |  |    |  |            |             |
|---|--|----|--|------------|-------------|
| 1 |  | 10 |  | 96.209999  | 610.280029  |
| 2 |  | 0  |  | 134.600006 | 415.450012  |
| 3 |  | 0  |  | 361.660004 | 1599.510010 |
| 4 |  | 0  |  | 22.139999  | 289.540009  |

|   | Customer_Status | Churn_Category  | Churn_Reason                  |
|---|-----------------|-----------------|-------------------------------|
| 0 | Stayed          | Others          | Others                        |
| 1 | Stayed          | Others          | Others                        |
| 2 | Churned         | Competitor      | Competitor had better devices |
| 3 | Churned         | Dissatisfaction | Product dissatisfaction       |
| 4 | Churned         | Dissatisfaction | Network reliability           |

[5 rows x 31 columns]

## Data Preprocessing

```
# List of columns to be label encoded
columns_to_encode = [
    'Gender', 'Married', 'City', 'Phone_Service', 'Multiple_Lines',
    'Internet_Service', 'Internet_Type', 'Online_Security', 'Online_Backup',
    'Device_Protection_Plan', 'Premium_Support', 'Streaming_TV', 'Streaming_Music',
    'Unlimited_Data', 'Contract', 'Paperless_Billing',
    'Payment_Method'
]

# Encode categorical variables except the target variable
label_encoders = {}
for column in columns_to_encode:
    label_encoders[column] = LabelEncoder()
    data[column] = label_encoders[column].fit_transform(data[column])

# Manually encode the target variable 'Customer_Status'
data['Customer_Status'] = data['Customer_Status'].map({'Stayed': 0, 'Churned': 1})

# Drop non-feature columns that are still strings or identifiers
data.drop(['Customer_ID', 'Churn_Category', 'Churn_Reason'], axis=1, inplace=True)

# Split data into features and target
X = data.drop('Customer_Status', axis=1)
y = data['Customer_Status']

# Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

## Train Random Forest Model

```
import pandas as pd
import numpy as np
```

```

import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, confusion_matrix
from sklearn.preprocessing import LabelEncoder
import joblib

# Initialize the Random Forest Classifier
rf_model = RandomForestClassifier(n_estimators=100, random_state=42)

# Train the model
rf_model.fit(X_train, y_train)

```

▼ **RandomForestClassifier** ⓘ ⓘ  
 RandomForestClassifier(random\_state=42)

## Evaluate Model

```

# Make predictions
y_pred = rf_model.predict(X_test)

# Evaluate the model
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# Feature Selection using Feature Importance
importances = rf_model.feature_importances_
indices = np.argsort(importances)[::-1]

# Plot the feature importances
plt.figure(figsize=(15, 6))
sns.barplot(x=importances[indices], y=X.columns[indices])
plt.title('Feature Importances')
plt.xlabel('Relative Importance')
plt.ylabel('Feature Names')
plt.show()

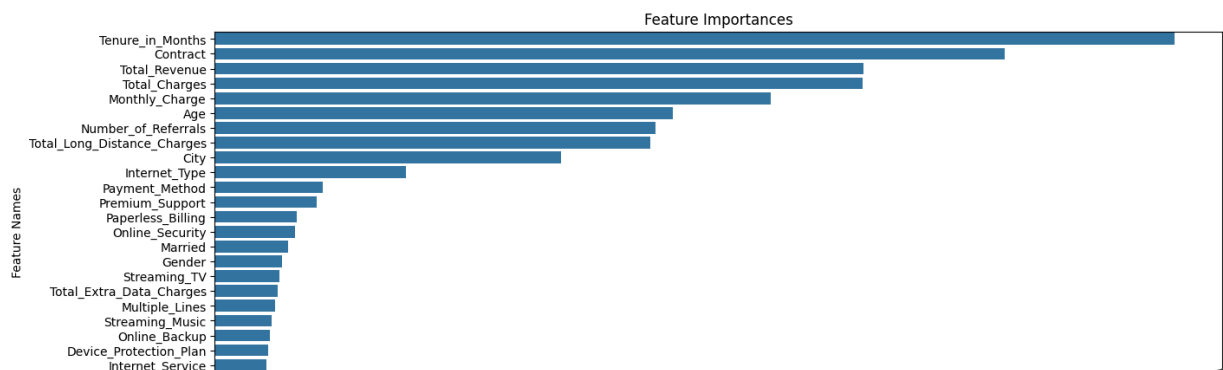
```

Confusion Matrix:

```
[[889  46]
 [136 247]]
```

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.87      | 0.95   | 0.91     | 935     |
| 1            | 0.84      | 0.64   | 0.73     | 383     |
| accuracy     |           |        | 0.86     | 1318    |
| macro avg    | 0.86      | 0.80   | 0.82     | 1318    |
| weighted avg | 0.86      | 0.86   | 0.86     | 1318    |



## Use Model for Prediction on New Data

```
path to the Joiner Data Excel file
'/content/sample_data/Prediction_data.xlsx'

sheet name to read data from
'vw_ChurnData'

data from the specified sheet into a pandas DataFrame
new_data = pd.read_excel(file_path, sheet_name=sheet_name)

first few rows of the fetched data
new_data.head()

original DataFrame to preserve unencoded columns
new_data = new_data.copy()

Customer_ID column
new_data['Customer_ID']

features that won't be used for prediction in the encoded DataFrame
new_data.drop(['Customer_ID', 'Customer_Status', 'Churn_Status'], axis=1)

encode categorical variables using the saved label encoders
for column in new_data.select_dtypes(include=['object']).columns:
    new_data[column] = label_encoders[column].transform(new_data[column])
```

```

original_data['Customer_Status_Predicted'] = new_predictions

# Create a new DataFrame to include only records predicted as "Churned"
churned_data = original_data[original_data['Customer_Status_Predicted'] == 'Churned']

# Save the results to a CSV file
churned_data.to_csv(r"/content/sample_data/Prediction.csv", index=False)

```

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